

Data Preparation for Machine Learning

Complete Lecture: From Raw Data to ML Pipeline

Department of Computer Science and Engineering

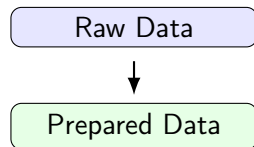
Complete Lecture Roadmap

- ① Foundations: why data preparation is essential
- ② Cleaning: missing values, duplicates, outliers
- ③ Transformation: scaling and encoding
- ④ Exploratory data analysis and feature understanding
- ⑤ Feature engineering and selection
- ⑥ End-to-end ML pipeline workflow

Learning Objectives

By the end of this lecture, students will be able to:

- Explain the role of data preparation in ML systems.
- Identify major data quality problems.
- Apply cleaning and transformation techniques.
- Understand missing values, duplicates, and outliers.
- Design a complete preprocessing workflow.



1. Role of Data Preparation in Machine Learning

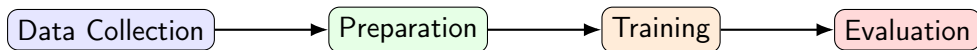
- Converts raw and noisy information into ML-ready features.
- Improves model accuracy, stability, and generalization.
- Reduces the effect of missing, inconsistent, and irrelevant data.
- Enables algorithms to discover meaningful patterns.

Key Idea

Better input data produces more reliable learning models.

Model Quality $\propto$ Data Quality

Machine Learning Workflow Overview



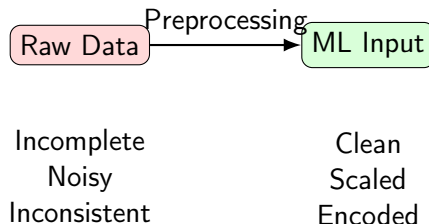
Focus of this lecture

Data cleaning, transformation, and feature preparation before model training.

Why Raw Data Cannot Directly Train ML Models

Common problems:

- Missing values
- Different feature scales
- Text and categorical attributes
- Noise and incorrect observations
- Duplicate records



Data Quality Problems

Problem	Example	Impact
Missing values	Empty fields	Information loss
Noise	Measurement errors	Wrong patterns
Duplicates	Repeated records	Biased learning
Inconsistency	Different formats	Low reliability
Outliers	Extreme values	Unstable models

Types of Data in ML

- Numerical data
- Categorical data
- Ordinal data
- Time-series data
- Text data
- Image data
- Audio data
- Sensor data

Types of Data in Machine Learning

Structured Data

- **Numerical Data**
 - Age = 25, Salary = \$50000
 - Temperature = 36.5°C
- **Categorical Data**
 - Gender = Male/Female
 - Country = Bangladesh, USA, Japan
- **Ordinal Data**
 - Rating = Poor, Good, Excellent
 - Education = School, Bachelor, Master
- **Time-Series Data**
 - Stock price over time
 - Daily temperature records

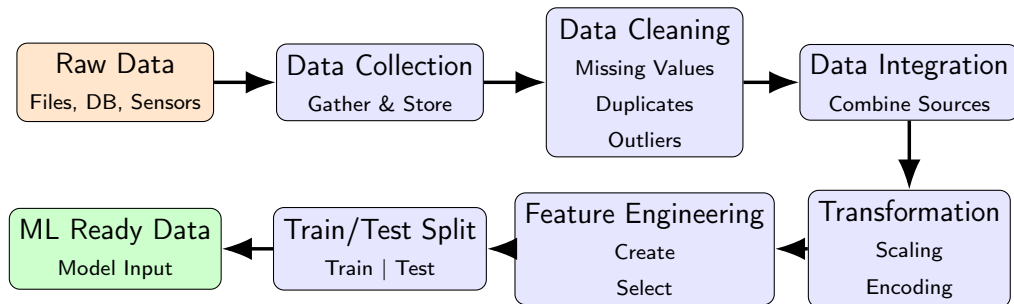
Unstructured / Complex Data

- **Text Data**
 - Customer reviews
 - News articles
 - Emails
- **Image Data**
 - X-ray images
 - Satellite images
- **Audio Data**
 - Speech recordings
 - Music signals
- **Sensor Data**
 - IoT temperature sensors
 - Vehicle speed sensors

Structured vs Unstructured Data

Type	Example	Storage	Processing
Structured	Tables, databases	Rows/columns	Scaling, encoding
Semi-structured	JSON, XML	Flexible schema	Parsing
Unstructured	Text, images	Files	Specialized methods

Complete Data Preprocessing Pipeline



Goal of Data Preprocessing

Transform raw, noisy, and inconsistent data into a clean, consistent, and meaningful dataset suitable for machine learning models.

Missing Values in Machine Learning

What are Missing Values?

Missing values represent unavailable or unknown information in a dataset.

Common Causes

- Data entry errors
- Sensor failures
- User omission
- Data integration problems

Example Dataset

Feature 1	Feature 2	Feature 3
5	10	Missing
7	12	9
8	Missing	11

Why Handle Missing Values?

Most ML algorithms cannot directly process incomplete data.

2. Missing Value Patterns

MCAR

Missing Completely At Random

- Missingness is random
- No relationship with data

Example:

A survey page fails randomly for some users.

MAR

Missing At Random

- Missingness depends on observed data
- Related variables are available

Example:

Income missing more often for younger users.

MNAR

Missing Not At Random

- Missingness depends on missing value itself
- Hidden information affects missingness

Example:

High earners avoid reporting income.

Importance of Missing Data Pattern

The missing data mechanism determines the appropriate handling strategy.

Missing Value Handling Techniques

Common Techniques

- | | |
|------------------------------------|--|
| 1. Deletion Methods | Remove rows/columns with many missing values |
| 2. Statistical Imputation | Mean, Median, Mode replacement |
| 3. Forward/Backward Filling | Use previous/next value (Time-series) |
| 4. Model-Based Imputation | Predict missing values using ML models |

Mean Imputation Example

Original : [5, 7, ?, 9]

$$\text{Mean} : \bar{x} = \frac{5 + 7 + 9}{3} = 7$$

After : [5, 7, 7, 9]

Selection Rule

Choose the method based on missing pattern, data type, and missing percentage.

Duplicate and Inconsistent Data

Duplicate Data

Duplicate records represent the same observation multiple times.

Example:

<i>ID</i>	<i>Name</i>	<i>Age</i>
101	John	25
101	John	25

Problem:

- Increases bias
- Distorts data distribution

Inconsistent Data

Same information stored in different formats.

Examples:

USA
U.S.A
United States

Solutions:

- Duplicate detection
- Data normalization
- Rule-based validation

Create a consistent, clean, and reliable dataset before model training.

Outlier Detection and Treatment

What is an Outlier?

An outlier is a data point that significantly differs from the normal pattern of a dataset.

Example:[20, 22, 21, 23, 95]

Detection Methods

- Statistical analysis
- Z-score method
- Interquartile Range (IQR)
- Visualization using box plots

Z-score Method

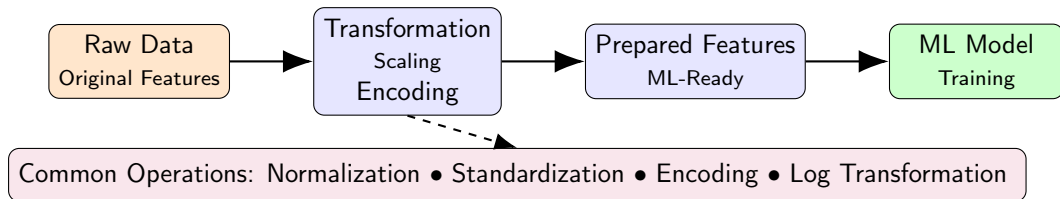
$$z = \frac{x - \mu}{\sigma}$$

- x = data value
- μ = mean
- σ = standard deviation

Rule of Thumb: $|z| > 3 \Rightarrow$ Possible Outlier

- Remove extreme values
- Transform data

3. Data Transformation in ML Pipeline



Purpose of Data Transformation

Convert raw variables into a suitable numerical representation that improves model learning and performance.

Why Data Transformation is Needed

Problems with Raw Data

Raw features may contain:

- Different feature scales
- Skewed distributions
- Extreme values (outliers)
- Inconsistent formats

Age = 25 Income = 500000

Different scales can affect model learning.

Benefits of Transformation

- **Improves numerical stability**
- **Reduces effect of extreme values**
- **Creates comparable feature scales**
- **Helps algorithms learn meaningful patterns**

Example

Before : [10, 100, 1000]

AfterScaling : [0.01, 0.10, 1.00]

Distribution Transformation Techniques

Log Transformation

$$x' = \log(x + 1)$$

Purpose:

- Handles positively skewed data
- Compresses large values
- Reduces impact of extreme values

Examples:

- Income
- Population
- Transaction amount

Power Transformation

$$x' = \frac{x^\lambda - 1}{\lambda}$$

Purpose:

- Changes feature distribution shape
- Makes data closer to normal distribution

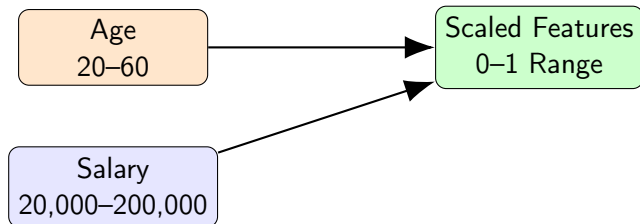
Common Methods:

- 1 Box-Cox Transformation
- 2 Yeo-Johnson Transformation

Why Transform Data?

Reduce skewness → Improve distribution →
Better model learning

Feature Scaling Concept



Why Feature Scaling?

Features with larger numerical ranges can dominate smaller features during model training.

Goal

Bring features to a comparable scale without changing their relationships.

Min-Max Normalization

Formula

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Characteristics

- Converts values into range [0, 1]
- Preserves relative distance between values
- Sensitive to outliers

Common Applications

Neural Networks • KNN • Distance-based Algorithms

Standardization (Z-score Scaling)

Formula

$$z = \frac{x - \mu}{\sigma}$$

where:

- μ = Mean
- σ = Standard deviation

Characteristics

- Mean becomes 0
- Standard deviation becomes 1
- No fixed range
- Less affected by scale differences

Common Applications

Linear Regression • Logistic Regression • SVM • PCA

Formula

$$x' = \frac{x - \text{median}(x)}{IQR}$$

Characteristics

- Uses median instead of mean
- Uses IQR instead of standard deviation
- Less sensitive to outliers

When to Use?

Datasets containing extreme values or heavy-tailed distributions.

Normalization vs Standardization

Aspect	Normalization	Standardization
Formula	Min-Max	Z-score
Output Range	0 to 1	No fixed range
Statistics Used	Minimum and Maximum	Mean and Standard Deviation
Outlier Effect	Highly sensitive	Less sensitive
Common Uses	Neural Networks, KNN	Linear Models, SVM, PCA

Selection Rule

Use Normalization for bounded features; use Standardization for statistical ML algorithms.

Label and Ordinal Encoding

Idea

Convert categories into numerical values.

Low = 0, *Medium* = 1, *High* = 2

When to Use?

- Categories have a natural order.
- Ranking information is meaningful.
- Common for ordinal features.

Examples

Education: High School < Bachelor < Master

Rating: Poor < Average < Excellent

Warning

Do not use for unordered categories because it creates false relationships.

One-Hot Encoding

Idea

Create separate binary features for each category.

$$\text{Red} = (1, 0, 0)$$

$$\text{Blue} = (0, 1, 0)$$

$$\text{Green} = (0, 0, 1)$$

Advantages

- No artificial ordering between categories.
- Works well for nominal data.
- Easy interpretation.

Example

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Target Encoding

Concept

Replace categories using statistics calculated from the target variable.

Example:

Category $\rightarrow$ *Average Target*

City_A $\rightarrow$ 0.75

City_B $\rightarrow$ 0.45

Applications

- High-cardinality features
- Many unique categories
- Recommendation systems
- Customer data

Important

Apply carefully to avoid target leakage.

4. Introduction to Exploratory Data Analysis (EDA)

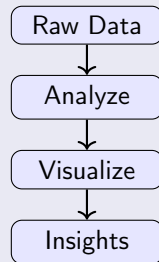
What is EDA?

Exploratory Data Analysis (EDA) is the process of analyzing and understanding data before applying machine learning models.

Main Objectives

- Discover hidden patterns
- Detect missing values and errors
- Understand feature distributions
- Identify relationships between variables

EDA Workflow



Statistical Analysis in EDA

Central Tendency

Describes the center of data distribution.

Mean, Median, Mode

- Mean: Average value
- Median: Middle value
- Mode: Most frequent value

Dispersion

Measures data variability.

Variance, Standard Deviation, Range

- Range: Data spread
- Variance: Overall variation
- Std. Dev.: Average deviation

Purpose in ML

Understand distribution, detect unusual values, and select suitable preprocessing methods.

Distribution Analysis

- **Histogram**
Shows feature distribution
- **Box Plot**
Detects outliers and spread

Relationship Analysis

- **Scatter Plot**
Shows variable relationships
- **Heatmap**
Displays correlations

Goal

Convert numerical patterns into visual insights.

Correlation Analysis

Correlation Coefficient

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Interpretation

- Measures linear relationship between variables
- Range: -1 to $+1$
- Helps identify related features

Example

Study Hours $\uparrow \Rightarrow$ Exam Score $\uparrow$
Positive correlation

Correlation Interpretation

Positive

$$r > 0$$

Variables increase together.

Example:

Temperature $\uparrow$ Ice cream sales $\uparrow$

Negative

$$r < 0$$

One increases while another decreases.

Example:

Price $\uparrow$ Demand $\downarrow$

No Linear Relation

$$r \approx 0$$

Weak linear relationship.

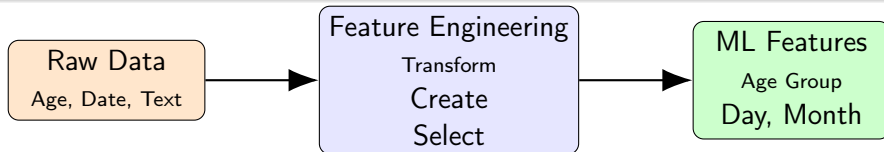
Important

Correlation shows association, not causation.

5. Introduction to Feature Engineering

Definition

Feature engineering is the process of transforming raw data into meaningful features that improve machine learning model performance.



Example

Date: 13-09-2026 → Day = Friday, Month = September

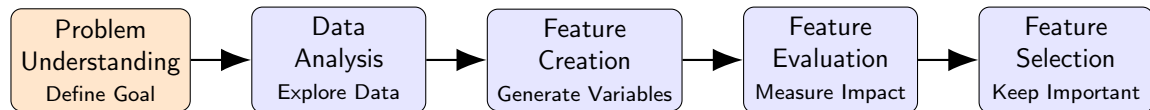
Why Feature Engineering is Important

- Improves predictive accuracy.
- Converts domain knowledge into data.
- Reveals hidden patterns.
- Reduces irrelevant information.

ML Principle

Better representation of data often produces better learning.

Feature Engineering Workflow



Feature Engineering Goal

Create informative features that improve model understanding while removing unnecessary information.

Feature Creation Techniques

1. Ratio Features

$$\text{Income Ratio} = \frac{\text{Income}}{\text{Family Size}}$$

2. Difference Features

$$\Delta X = X_{\text{current}} - X_{\text{previous}}$$

3. Aggregation Features

Mean, Max, Min, Count

Interaction Features

Combine multiple variables to capture hidden relationships.

Example:

$$\text{Area} \times \text{Price}$$

$$\text{Height} \times \text{Weight}$$

Purpose:

Capture complex patterns that individual features cannot represent.

Create informative numerical representations from existing variables.

Date-Time Feature Engineering

Raw Date-Time

2026 – 09 – 11

17 : 30 : 00

Generated Features

Date Features Day, Month, Year

Calendar Features Weekend Flag, Holiday

Season Features Winter, Summer, Rainy

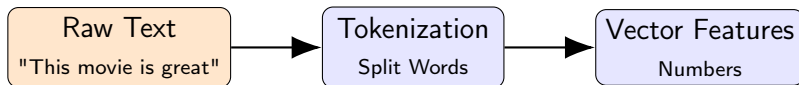
Time Features Hour, Time Period

Timestamp



ML Features
Day, Month, Hour

Text Feature Engineering



Common Techniques

- **Bag of Words**
Word frequency representation
- **TF-IDF**
Importance of words
- **Word Embeddings**
Semantic meaning representation
- **Text Statistics**
Length, count, frequency

Example

Text:

"Machine learning is powerful"

Features:

[0.25, 0.81, 0.42, ...]

Ready for ML models

Domain-Based Feature Creation

Feature Engineering Uses Domain Knowledge

Expert knowledge helps transform raw information into meaningful predictive features.

Domain	Raw Data	Created Feature
Finance	Transactions	Spending pattern, Credit score
Healthcare	Patient measurements	Risk score, Health indicator
Retail	Customer purchases	Customer lifetime value
Education	Student activity	Performance indicator

Key Idea

Raw data → Domain Knowledge → Useful Features

Feature Selection vs Feature Extraction

Feature Selection

Idea: Select important features from original data.

$$X_1, X_2, X_3, X_4 \implies X_1, X_3$$

Characteristics:

- Keeps original features
- Reduces irrelevant variables
- Improves efficiency

Examples: Correlation, RFE, LASSO

Feature Extraction

Idea: Create new features from existing variables.

$$X_1, X_2, X_3, X_4 \implies Z_1, Z_2$$

Characteristics:

- Creates new representation
- Reduces dimensionality
- May lose original meaning

Examples: PCA, Autoencoders, Projection

Key Difference

Feature Selection: Choose existing features

Feature Extraction: Create new features

Feature Selection Methods

Filter Methods

Select features using statistical measures.

- Correlation
- Chi-square test
- Mutual information

Wrapper Methods

Evaluate feature subsets using models.

- Recursive Feature Elimination (RFE)
- Forward Selection
- Backward Elimination

Embedded Methods

Selection occurs during model training.

- LASSO Regression
- Decision Trees
- Random Forest

Goal of Feature Selection

Reduce dimensionality → Remove irrelevant features → Improve model efficiency

Feature Engineering Case Study: Customer Analytics

Objective

Transform customer activity data into meaningful features for prediction and decision-making.

Original Data	Engineered Feature
Purchase History	Purchase Frequency
Order Amount	Average Order Value
Customer Activity	Customer Lifetime Value
Transaction Dates	Recency of Purchase

Feature Engineering Idea

Raw Customer Data → Meaningful Business Features → ML Model

Feature Engineering: Best Practices and Challenges

Best Practices

- Use domain knowledge
- Keep features interpretable
- Validate feature usefulness
- Avoid data leakage
- Maintain feature consistency

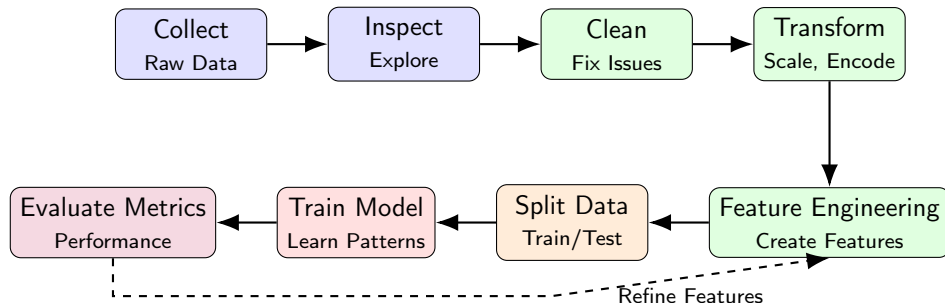
Challenges

- Too many features
- Overfitting risk
- High computational cost
- Difficult feature maintenance
- Data quality issues

Goal of Feature Engineering

Create fewer but more informative features that improve model performance.

6. End-to-End Data Preparation Workflow



Key Idea

Data preparation is an iterative process where model evaluation provides feedback to improve feature quality and preprocessing decisions.

- Data preparation determines the quality of machine learning systems.
- A complete workflow combines cleaning, transformation, analysis, engineering, and validation.
- Proper preprocessing improves accuracy, reliability, and generalization.

Thank You

Questions?